Wi-Fi (WLAN) Compliance Testing

Before going through these steps refer to the appendix **4.2_Requirements for the WiFi Checklist**
This testing needs to be completed before implementation. Wi-Fi testing must be done along the path of the operation of the AGV. The device needed for testing is a laptop.

Step 1: Find out the IP address of the WiFi network

Step 2: Open Terminal (macbook)/ Command Prompt (Windows)

Step 3: Use the command: **ping 192.168.100.89** ← This is an example IP address of the Wi-Fi.
Use IP address of the Wi-Fi to which AGV and server will be connected to.

Step 4: Then check the results and see if there is any data packet loss

```
C:\Users\ArnabMitra>ping 192.168.100.89

Pinging 192.168.100.89 with 32 bytes of data:
Reply from 192.168.100.89: bytes=32 time=236ms TTL=128
Reply from 192.168.100.89: bytes=32 time=3ms TTL=128
Reply from 192.168.100.89: bytes=32 time=5ms TTL=128
Reply from 192.168.100.89: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.100.89:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 236ms, Average = 61ms
```

Step 5: If there is any loss of data packet then identify if there is external source which is causing interference and determine how to resolve this issue.

Step 6: Test MUST be conducted along the intended path of travel/operation.

Step 7: Test must be conducted every 5 meters to get a better understanding of the network coverage.